Regular expressions that solve everyday text problems

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START HERE

The problem comes first.

A regex is useful when the rule is clearer than a pile of string operations.



Find

Locate a substring.



Validate

Check a whole input.



Extract

Turn text into fields.



Replace

Rewrite a matched form.



Split

Separate a repeated structure.

```
Think of regex as a small language for describing sets of possible strings.
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THE SMALL VOCABULARY

Build patterns from a few reliable primitives

Most everyday patterns are assembled, not invented.

What may appear

`[A-Z]` uppercase letters
`[0-9]` digits
`[^,]` anything except a comma

How often

`?` optional · `*` zero or more
`+` one or more · `{m,n}` bounded

Where it sits

`^` start · `$` end
`\b` word boundary

How it branches

`(…)` capture · `(?:…)` group
`|` choose between alternatives

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EXTRACTION

Extract structure from messy text

Turn logs, tickets, filenames, and copied text into fields you can analyze.

Use named or numbered groups to extract an issue key, date, and severity from one line; treat matches as structured data, not just highlighted text.

INPUT

[SEV:high] INC-4821 · 2026-10-22

Field	Value
key	INC-4821
date	2026-10-22
severity	high



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CANONICAL FORM

Normalize before you compare

01

Match

Accept the variants you expect.

02

Capture

Keep the meaningful value.

03

Rewrite

Replace separators, spaces, or casing.

04

Test

Use representative examples.

" ACME — 42 " → "acme-42"

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VALIDATION

Boundaries make the contract visible

Before writing syntax, decide what counts as a valid input.

Question	Contract	Pattern consequence
Whole input or substring?	Validate the complete value.	^…$ or the engine’s full-match API
What is optional?	State the allowed variants.	Group the optional part with ?
What belongs in code?	Keep complex business rules explainable.	Regex checks shape; ordinary code checks policy

Regex can validate a predefined shape. When the expression becomes harder to understand than the rule, switch tools.

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DEBUGGING

Debug patterns like small programs

A failing example is more useful than a clever rewrite.

Case	Example	What it reveals
✓ Positive	INC-4821	The intended shape matches.
✗ Negative	INC_4821	An invalid separator is rejected.
Boundary	XINC-4821Y	Anchors or word boundaries behave as intended.
⌀ Empty	""	Optional pieces do not accidentally accept nothing.

Build incrementally

Start with the literal. Add one character class, quantifier, or group at a time.

Check the engine

Python, JavaScript, .NET, and shell tools do not share every feature or default.

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THE REUSABLE LOOP

A reusable regex workflow beats clever syntax

1

Contract

State the input contract and desired output.

2

Generalize

Start literal; generalize only where examples demand it.

3

Readability

Keep patterns readable with names, comments, and tests.

4

Switch tools

When a pattern is harder to maintain than the text logic, switch tools.

Readable patterns are named, commented, and tested.

The goal is dependable text logic, not the smallest possible pattern.

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Geo Tag



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Thank You

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